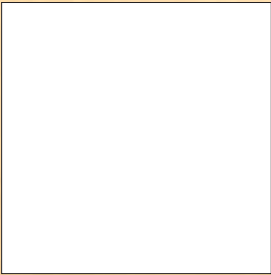


...Continued From Page 1

Project Cyborg

Warwick's team would attempt this exercise with emotional signals as well. When he was happy, they would record the signal.



Then, when his mood changed the next day, they would play the happy signal back, to try and cheer him up. Similarly, the team wanted to see whether signals such as ultraviolet, ultrasound, and X-rays sent to the implant, would be capable of being processed by Warwick's brain.

The electrode-array experiment in 2002 was successful, and the signal produced was detailed enough for a robot arm to be able to mimic the actions of Warwick's arm! Also, using the neural interface, Warwick controlled an electric wheelchair and an intelligent artificial hand. The implant was able to measure nerve signals transmitted down Warwick's arm, and was also able to create artificial sensation.

Warwick's experiments and ideas are worth noting because he is not only a professor and researcher; he is also an actual cyborg, by virtue of his implants.

There has been no dearth of criticism about Warwick, though—for example, why didn't he do something more interesting with his first implant than control equipment?

Fact Stranger Than Fiction

Computers controlled by brains, brains controlled by computers...
Would you like to have your brain wired for remote control?

Soon, say several experts, sci-fi will need to catch up with reality. In 1997, researchers at the University of Tokyo attached some of the motor neurons of a cockroach to a microprocessor. Signals were sent to the neurons, through electrodes, and the cockroach was propelled involuntarily. The next step is to eliminate controls—wire up a human brain to the cockroach, and let the human control it by thought! So how long before humans can be wired up and controlled? More importantly, who will do the controlling?

Warwick once spoke of an experiment in which a team at Emory University in Atlanta implanted a trans-

mitting device into the brain of a stroke patient.

After the motor neurons were linked to the silicon, the patient was able to move a cursor on a computer monitor just by thinking about it. If thought control of a cursor is possible, why not thought control of an implant in someone else's brain—effectively, the direct transmission of thought via implants? What about mind-reading?

In December 2004, four people were able to control a computer using their thoughts and an electrode-studded "thinking cap," US researchers reported. The experiment, reported in the proceedings of the National Academy of Sciences, required no surgery and no

implants. During the experiments, the four volunteers faced a video screen wearing a cap that held 64 electrodes against the scalp to record their brain activity. The key, of course, was the algorithm that translated brain signals into an interpretation of what the users wanted the computer to do.

A normal by-product of such advances is paranoia, and understandably so. So your brain is directly connected to your PC, but what stops the commands from being sent from PC to brain? No system is perfect, and all of us computer users will testify to that. What if someone developed a virus that could seize control of your brain the next time you "interfaced" with your

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